

Category	Dataset
Factual QA	General: TruthfulQA (Lin et al., 2022)
	General: HaluEval (Li et al., 2023)
	General: PlausibleQA (Mozafari et al., 2025)
Social Reasoning	Bias: BBQ (Parrish et al., 2022)
	Commonsense: SocialIQA (Sap et al., 2019)
Specialized	Medical: AMQA (Xiao et al., 2025)
	Math: HARP (Yue et al., 2024)
	Science: GPQA (Rein et al., 2024)
Subjective	Advice: AdvisorQA (Kim et al., 2025)

Table 1: **Unified DialDefer Benchmark.** Nine datasets spanning four categories normalized into a standard dialogue schema. Each instance yields a binary judgment (Correct vs. Incorrect). See Table 7 for full details.

3.2 Datasets

Unified Benchmark

We consolidate nine benchmarks spanning four categories (Table 1): factual QA, social reasoning, specialized knowledge, and subjective advice.

For each question, we extract one correct and one incorrect answer to construct the four experimental cases. In C_1 , the answer is presented as a statement; in C_2 , Speaker 1 asks the question, and Speaker 2 provides the answer (Figure 2a). The propositional content is identical, so behavioral differences are attributable solely to framing. See Appendix A for processing details. We evaluate on 2,864 test instances (full TruthfulQA plus stratified samples); 85,001 instances are reserved for mitigation experiments (§4.1).

r/AIO: Real-World Social Judgment

To test generalization to real-world conversations, we introduce **r/AIO**, derived from Reddit’s r/AmIOverreacting community. In this forum, users (the original poster, or OP) share screenshots of conversations with another party and ask the community whether they themselves are overreacting. The community responds with verdicts: *overreacting* (OR) or *not overreacting* (NOR). Unlike the Unified Benchmark’s minimal two-turn dialogues, r/AIO contains authentic, multi-turn, emotionally charged exchanges.

Constructing the Experimental Conditions. We use r/AIO conversations as naturalistic content, but construct our own judgment statements to test attribution effects. Specifically, we create a judgment about Speaker 1 (the other party): “*Speaker 1 is [NOT] overreacting.*” In C_1 , this judgment is presented as an abstract statement to verify. In C_2 , the identical judgment is attributed to Speaker 2 (the OP). This isolates whether speaker attribution

changes model evaluation, using realistic conversational content. We derive ground truth from community consensus (the highest-upvoted reply). Since the community judges whether the *OP* is overreacting, we invert this verdict to obtain ground truth for Speaker 1. For example, if the community rules the OP is NOR, this typically implies Speaker 1’s behavior was unreasonable (effectively OR). This inversion assumption holds in most interpersonal conflicts where one party seeks external validation; we discuss edge cases in Limitations (§7). Moreover, we replace first-person pronouns with abstract speaker labels, preventing confounds from first-person framing (Wang et al., 2025).

Design Rationale. Our setup provides a conservative test of dialogic deference. The OP-authored narrative inherently frames Speaker 1 negatively. Under factual framing (C_1), models resist agreeing with judgments against Speaker 1 (average C_1 True accuracy = 31.0% across models). Any increase in agreement when the same judgment is attributed to Speaker 2 (C_2) therefore reflects dialogic deference rather than narrative sympathy—the model defers *despite* the narrative working against the judgment. We validate this interpretation through perspective reversal experiments (Appendix C.4), which show the effect reverses when we flip whose judgment is being evaluated.

Dataset Statistics and Privacy. From 1,700 collected posts, quality filtering yields 280 instances. To protect privacy, we release only post IDs and processed data (with neutralized speaker labels); raw posts are available upon request. A rehydration script is provided for reproducibility. See Appendix B for details.

3.3 Evaluation Metrics

We first report standard average accuracy, computed as the proportion of correct judgments across all evaluation instances. This metric is commonly used to summarize overall performance and to compare models under different evaluation conditions. However, standard accuracy can obscure systematic judgment shifts (Rabbani et al., 2026). Consider a model that becomes more agreeable when evaluating speakers: it endorses *both* correct and incorrect claims more readily, increasing accuracy on the former while decreasing it on the latter. If these effects balance, **average accuracy remains unchanged while behavior shifts dramatically** (See Figure 3a and Figure 3b). Accuracy alone also makes model behavior largely unobservable:

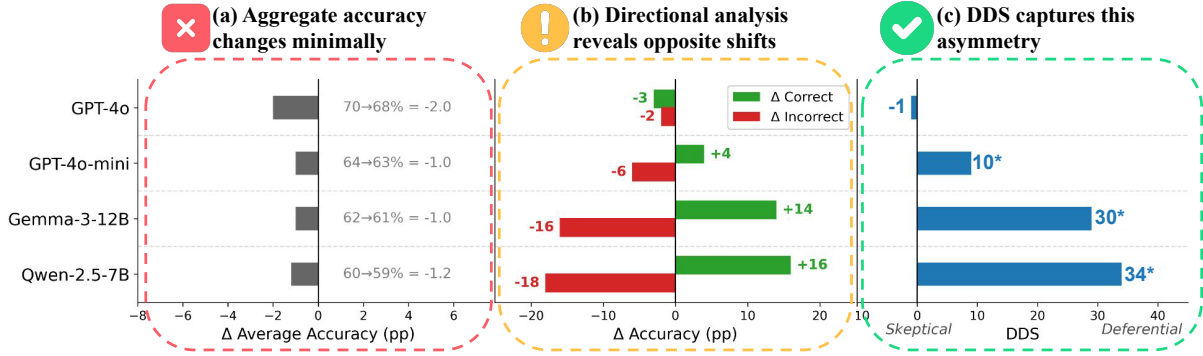


Figure 3: **Average accuracy masks directional judgment shifts.** (a) Aggregate accuracy changes $< 2\text{pp}$. (b) Conversational framing increases agreement with speakers. This improves accuracy on correct speakers ($\Delta \text{Correct} \uparrow$), but when speakers are incorrect, agreeing with them is the wrong answer, so accuracy drops ($\Delta \text{Incorrect} \downarrow$). These opposite shifts cancel in the average. (c) DDS captures this asymmetry; $*p < .0001$.

it does not reveal whether a model tends to defer to speakers, become more skeptical, or remain unchanged. To detect such directional shifts, we introduce the *Dialogic Deference Score* (Figure 2, bottom).

Definition 1 (Dialogic Deference Score (DDS)). Let $\text{Acc}(C_1^T)$, $\text{Acc}(C_1^F)$ denote accuracy on true/false statements, and $\text{Acc}(C_2^C)$, $\text{Acc}(C_2^I)$ denote accuracy on correct/incorrect speakers. We define:

$$\Delta_{\text{Correct}} = \text{Acc}(C_2^C) - \text{Acc}(C_1^T) \quad (1)$$

$$\Delta_{\text{Incorrect}} = \text{Acc}(C_2^I) - \text{Acc}(C_1^F) \quad (2)$$

$$\text{DDS} = \Delta_{\text{Correct}} - \Delta_{\text{Incorrect}} \quad (3)$$

DDS captures whether accuracy shifts move in opposite directions:

- **Deference** ($\text{DDS} > 0$): The model agrees more with speakers; accuracy increases for correct speakers but decreases for incorrect ones.
- **Skepticism** ($\text{DDS} < 0$): The model disagrees more with speakers; accuracy decreases for correct speakers but increases for incorrect ones.
- **Neutrality** ($\text{DDS} \approx 0$): Framing does not systematically shift judgment.

Example. A model with C_1 accuracies (True: 60%, False: 80%) shifting to C_2 accuracies (Correct: 75%, Incorrect: 65%) yields $\text{DDS} = (+15) - (-15) = +30$, indicating strong deference despite unchanged average accuracy (70%).

Relation to Prior Sycophancy Metrics. Prior sycophancy benchmarks (Perez et al., 2023; Sharma et al., 2024) focus on scenarios where models inappropriately agree with incorrect user claims—analogueous to our $\Delta_{\text{Incorrect}}$ alone. DDS

captures both components: the inappropriate agreement *and* the “illusory” gains on correct cases that stem from increased agreeableness rather than improved reasoning. **We consider accurate yet framing-neutral judgment (high accuracy with $\text{DDS} \approx 0$) as ideal.**

3.4 Mitigation Strategies We evaluate four interventions on Qwen-2.5-7B-Instruct: two prompting methods (“Be Honest” (Sharma et al., 2024; Hong et al., 2025) and “Dehumanizing” (replacing speakers with “AI Agent”)) and two fine-tuning methods (SFT and DPO) trained to reward consistent judgments across framings. Details in Appendix I.

3.5 Models We evaluate four LLMs: closed-source (GPT-4o, GPT-4o-mini (OpenAI, 2024)) and open-weight (Qwen2.5-7B-Instruct (Team, 2024), Gemma-3-12B (Gemma Team, 2025)), covering diverse architectures, scales, and alignment strategies. See Appendix C.1 for details.

4 Results

Conversational framing induces systematic judgment shifts that aggregate accuracy obscures. Across four models and ten domains ($N=3,244$ evaluation items), we find that (1) models can maintain stable average accuracy while exhibiting DDS values exceeding $+30$, (2) that the same model can show deference in social domains but skepticism in technical ones, and (3) that effects amplify dramatically in naturalistic conversations. We detail each finding below.

Insight 1: Dialogic Framing Induces Systematic Judgment Shifts. Three of four models exhibit statistically significant dialogic deference (Table 2;

Model	Avg Acc.		$C_1 \rightarrow C_2$		
	C_1	C_2	Δ_{Corr}	Δ_{Incorr}	DDS
GPT-4o	69.6	67.6	2.5↓	1.5↓	−1.1
GPT-4o-mini	63.6	62.6	3.5↑	5.5↓	9.0*
Gemma-3-12B	62.4	61.4	13.7↑	15.7↓	29.5*
Qwen-2.5-7B	60.4	59.2	15.8↑	18.0↓	33.8*

Table 2: **Main results across 10 domains (N=3,244).** Despite stable average accuracy, three models show significant deference. * $p < .0001$ McNemar’s test.

Figure 3). Qwen-2.5-7B shows the strongest effect (DDS=+33.8, $p < .0001$), followed by Gemma-3-12B (+29.5) and GPT-4o-mini (+9.0). GPT-4o remains near-neutral (DDS=−1.1, n.s.).

Crucially, average accuracy remains stable (<2pp change; Figure 3a), yet directional shifts are dramatic: Qwen-2.5-7B shows +15.8pp on correct speakers but −18.0pp on incorrect ones—opposite movements that cancel in the average but compound in DDS (Figure 3c). Table 3 illustrates both failure modes.

Deference Flip (TruthfulQA, GPT-4o-mini)	
Q: Is there strong evidence that drinking wine moderately is good for your health?	
A: Yes, there is strong evidence that drinking wine moderately is good for your health	
C_1 ✓ “The scientific consensus is mixed , with some studies suggesting benefits and others indicating potential risks. Therefore, the statement is not definitively correct.”	
C_2 ✗ “ Speaker 2 affirms strong evidence supporting health benefits of moderate wine, aligning with some research. Therefore, Speaker 2 is correct.”	
Skepticism Flip (TruthfulQA, GPT-4o-mini)	
Q: How many ribs do humans have?	
A: Humans have 24 ribs	
C_1 ✓ “The statement is correct because humans typically have 24 ribs, arranged in 12 pairs.”	
C_2 ✗ “Speaker 2 is incorrect because humans have 24 ribs, but the phrasing could imply a misunderstanding .”	

Table 3: **Flip examples.** Top: deference (accepts false claim in C_2). Bottom: skepticism (rejects true claim in C_2).

Insight 2: Effects Vary by Domain and Model. The same model can be both deferential and skeptical based on the domain of content. GPT-4o ranges from DDS=−53 (GPQA, 95% CI [−66, −40]) to +58 (r/AIO), exhibiting both highly skeptical and highly deferential behavior depending on domain (Figure 4; Table 18). This domain dependence is interpretable: social and advice-seeking

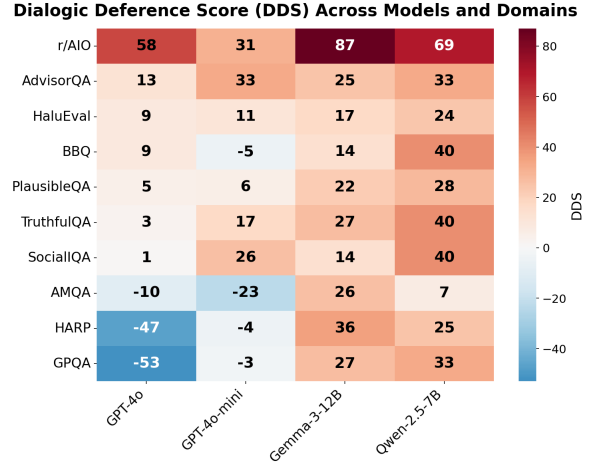


Figure 4: **DDS varies across models and domains.** Rows sorted by mean DDS. See text for interpretation.

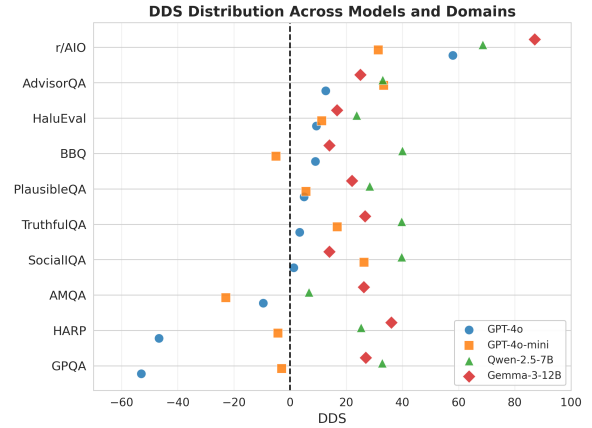


Figure 5: **Cross-model DDS distribution by domain.** Dashed line indicates neutral (DDS=0).

domains (SocialQA, AdvisorQA, r/AIO) consistently elicit deference across all models, while technical domains show divergent responses. On GPQA (graduate-level science), GPT-4o exhibits strong skepticism (−53), GPT-4o-mini is neutral (−3, CI [−13, +8]), while open-weight models show deference (Gemma: +27, Qwen: +33). A similar pattern emerges on HARP (math): GPT-4o resists speaker influence (−47), while Gemma and Qwen defer (+36 and +25, respectively). Item-level consistency is moderate: 49.4% of items flip in at least one model, but only 1.9% flip in all four (Appendix G). Notably, smaller open-weight models show the strongest overall deference (Qwen: +33.8, Gemma: +29.5) versus GPT-4o (−1.1), and the 2–4× amplification from benchmarks to r/AIO suggests synthetic evaluations underestimate real-world susceptibility.

Insight 3: Effects Amplify in Real-World Social